Brutal Series - Paper B59 (Transport | Heat | Data Handling | Symmetry)

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A boy's heart beats 72 times a minute at rest. During fast running the rate goes up by five-sixths of the resting rate. How many times does his heart beat in 4 minutes of running? [Transportation in Animals & Plants]

- **(A)** 264
- **(B)** 480
- **(C)** 288
- **(D)** 528
- **(E)** 240
- **(F)** 396
- **(G)** 132
- **(H)** 552

2. The mean of 6 numbers is 24. If the number 39 is removed from the set, what is the mean of the remaining 5 numbers? [Data Handling]

- **(A)** 105
- **(B)** 24
- **(C)** 20
- **(D)** 19
- **(E)** 21
- **(F)** 18

3. In a lab, water at 24 C is heated so its temperature rises 4 C every minute for 6 minutes. Then it is cooled and loses 3 C every minute for 2 minutes. What is the final temperature? [Heat]

- **(A)** 36 C
- **(B)** 30 C
- **(C)** 54 C
- **(D)** 18 C
- **(E)** 42 C
- **(F)** 48 C
- **(G)** 44 C

4. A regular polygon has 9 lines of symmetry. What is the smallest angle through which it can be rotated to look exactly the same? [Symmetry]

- **(A)** 80 degrees
- **(B)** 9 degrees
- **(C)** 90 degrees
- **(D)** 45 degrees
- **(E)** 40 degrees
- **(F)** 36 degrees
- **(G)** 60 degrees

5. A nurse feels the pulse and counts 26 beats in 20 seconds. What is the pulse rate per minute? [Transportation in Animals & Plants]

- **(A)** 13
- **(B)** 78
- **(C)** 98
- **(D)** 156
- **(E)** 104
- **(F)** 26
- **(G)** 52

6. Rainfall (in mm) over 7 days was 5, 7, 9, 9, 9, 12, 12. What is the range plus the mode of this data? [Data Handling]

- **(A)** 12
- **(B)** 7
- **(C)** 18
- **(D)** 9
- **(E)** 21
- **(F)** 16
- **(G)** 2

7. A clinical thermometer can read between 35 C and 42 C. A patient's reading is 39.4 C. By how much is this above the normal body temperature of 37 C? [Heat]

- **(A)** 1.6 C
- **(B)** 2.2 C
- **(C)** 39.4 C
- **(D)** 4.4 C
- **(E)** 2.4 C
- **(F)** 5 C

8. A figure has rotational symmetry of order 6. What is the smallest angle of rotation that maps it onto itself? [Symmetry]

- **(A)** 6 degrees
- **(B)** 30 degrees
- **(C)** 120 degrees
- **(D)** 72 degrees
- **(E)** 60 degrees
- **(F)** 360 degrees
- **(G)** 45 degrees

9. An adult has about 5 litres of blood. Red blood cells make up 45% of this volume. How many millilitres are NOT red blood cells (plasma and other cells)? [Transportation in Animals & Plants]

- **(A)** 3250 mL
- **(B)** 2755 mL
- **(C)** 5000 mL
- **(D)** 500 mL
- **(E)** 2750 mL
- **(F)** 275 mL

10. Take the first five multiples of 4, add them up, and find their average value. [Data Handling]

(A) 10
(B) 11
(C) 60
(D) 14
(E) 4
(F) 12
(G) 20

11. Two identical mugs of water are left in the sun. The black mug's temperature rises 12 C; the white mug's rises only one-third as much. What is the difference between the two temperature rises? [Heat]

(A) 9 C
(B) 4 C
(C) 16 C
(D) 12 C
(E) 3 C
(F) 6 C
(G) 8 C

12. For a square, multiply the number of lines of symmetry by its order of rotational symmetry. What is the product? [Symmetry]

(A) 2
(B) 12
(C) 8
(D) 24
(E) 20
(F) 16
(G) 4

13. A plant's roots absorb 200 mL of water in a day. Of this, 90% is lost from the leaves by transpiration. How much water is kept by the plant that day? [Transportation in Animals & Plants]

(A) 180 mL
(B) 200 mL
(C) 2 mL
(D) 100 mL
(E) 20 mL
(F) 18 mL

14. Five students scored 60, 70, 80, 90 and 55 in Maths. What is their mean score? [Data Handling]

(A) 71
(B) 72
(C) 75
(D) 355
(E) 69
(F) 70
(G) 65

15. A heater raises the temperature of water at a steady 5 C per minute. How many minutes does it take to heat the water from 20 C to 80 C? [Heat]

- **(A)** 4 minutes
- **(B)** 12 minutes
- **(C)** 16 minutes
- **(D)** 6 minutes
- **(E)** 15 minutes
- **(F)** 10 minutes
- **(G)** 60 minutes

16. How many lines of symmetry does an equilateral triangle and a rectangle have IN TOTAL? [Symmetry]

- **(A)** 8
- **(B)** 2
- **(C)** 7
- **(D)** 4
- **(E)** 6
- **(F)** 3
- **(G)** 5

17. The heart pushes out 70 mL of blood with each beat and beats 72 times a minute. How much blood does it pump in one minute, in litres? [Transportation in Animals & Plants]

- **(A)** 4.9 L
- **(B)** 504 L
- **(C)** 0.504 L
- **(D)** 5.4 L
- **(E)** 5040 L
- **(F)** 50.4 L
- **(G)** 5.04 L

18. A bag has 3 red, 5 blue and 2 green balls. One ball is drawn at random. What is the probability that it is NOT blue? [Data Handling]

- **(A)** 4/5
- **(B)** 1/2
- **(C)** 1/10
- **(D)** 1/5
- **(E)** 7/10
- **(F)** 3/10

19. At a coast the sea breeze blows for 11 hours and the land breeze for 9 hours in a 24-hour day; the rest of the time is calm. How many hours are calm? [Heat]

- **(A)** 4 hours
- **(B)** 20 hours
- **(C)** 3 hours
- **(D)** 6 hours
- **(E)** 0 hours
- **(F)** 5 hours
- **(G)** 2 hours

20. A figure looks exactly the same after a rotation of 24 degrees (its smallest such angle). What is its order of rotational symmetry? [Symmetry]

- **(A)** 336
- **(B)** 20
- **(C)** 10
- **(D)** 12
- **(E)** 8
- **(F)** 24
- **(G)** 15

21. A dialysis machine cleans the blood of a patient whose blood urea is 0.42 g per litre. After the session the urea falls to one-sixth of this. What is the new urea level? [Transportation in Animals & Plants]

- **(A)** 0.36 g/L
- **(B)** 2.52 g/L
- **(C)** 0.07 g/L
- **(D)** 0.06 g/L
- **(E)** 0.35 g/L
- **(F)** 0.7 g/L
- **(G)** 0.252 g/L

22. The mean of 8 observations is 15. If each observation is increased by 3, what is the new mean? [Data Handling]

- **(A)** 120
- **(B)** 21
- **(C)** 16
- **(D)** 24
- **(E)** 18
- **(F)** 15
- **(G)** 45

23. A teacher lists 8 objects; 5 of them are metals (good conductors) and the rest are insulators. What fraction of the objects are insulators? [Heat]

- **(A)** 3/13
- **(B)** 3/5
- **(C)** 3/8
- **(D)** 1/2
- **(E)** 8/3
- **(F)** 5/8
- **(G)** 5/3

24. For a regular hexagon, add its number of lines of symmetry to its order of rotational symmetry. What do you get? [Symmetry]

- **(A)** 6
- **(B)** 12
- **(C)** 10
- **(D)** 8
- **(E)** 24
- **(F)** 18
- **(G)** 36

25. Blood has about 5,000,000 red blood cells in every microlitre. How many red blood cells are there in 3 microlitres of the same blood? [Transportation in Animals & Plants]

- **(A)** 50,000,000
- **(B)** 15,000,000
- **(C)** 45,000,000
- **(D)** 150,000,000
- **(E)** 1,500,000
- **(F)** 16,666,666

26. A bar graph shows trees planted in 4 years as 120, 150, 90 and an unknown number. If the mean over the 4 years is 130, how many trees were planted in the fourth year? [Data Handling]

- **(A)** 150
- **(B)** 520
- **(C)** 120
- **(D)** 160
- **(E)** 130
- **(F)** 100
- **(G)** 140

27. In one day a town's temperature was 18 C in the morning, 33 C at noon and 12 C at night. What was the temperature range for that day? [Heat]

- **(A)** 21 C
- **(B)** 33 C
- **(C)** 18 C
- **(D)** 6 C
- **(E)** 9 C
- **(F)** 15 C

28. Among the letters H, O, S, T and X, how many have rotational symmetry of order 2 (look the same after a half turn)? [Symmetry]

- **(A)** 0
- **(B)** 2
- **(C)** 1
- **(D)** 3
- **(E)** 6
- **(F)** 4
- **(G)** 5

29. A child's pulse is 90 beats per minute. If the pulse is counted for only 40 seconds, how many beats are counted? [Transportation in Animals & Plants]

- **(A)** 90
- **(B)** 36
- **(C)** 120
- **(D)** 60
- **(E)** 54
- **(F)** 135
- **(G)** 45

30. Find the median of the data: 12, 15, 18, 20, 22, 25. [Data Handling]

- **(A)** 21
- **(B)** 17
- **(C)** 19.5
- **(D)** 19
- **(E)** 18
- **(F)** 18.5
- **(G)** 20

31. A metal rod 80 cm long expands 0.6 mm for every 10 C rise in temperature. How much will it expand if its temperature rises by 50 C? [Heat]

- **(A)** 30 mm
- **(B)** 0.3 mm
- **(C)** 2.5 mm
- **(D)** 6 mm
- **(E)** 3 mm
- **(F)** 15 mm

32. A figure maps onto itself every 45 degrees of rotation. How many distinct positions (including the starting one) does it have in a full turn? [Symmetry]

- **(A)** 6
- **(B)** 8
- **(C)** 4
- **(D)** 45
- **(E)** 16
- **(F)** 9

33. While playing, a child loses 1.2 litres of sweat in 3 hours. What is the average rate of sweat loss in millilitres per hour? [Transportation in Animals & Plants]

- **(A)** 4000 mL/hour
- **(B)** 400 mL/hour
- **(C)** 360 mL/hour
- **(D)** 300 mL/hour
- **(E)** 40 mL/hour
- **(F)** 1200 mL/hour
- **(G)** 0.4 mL/hour

34. A fair die is rolled once. What is the probability of getting a prime number? [Data Handling]

- **(A)** 2/3
- **(B)** 1/2
- **(C)** 1/3
- **(D)** 5/6
- **(E)** 1/4
- **(F)** 1/6
- **(G)** 4/6

35. Three thin blankets each trap a 2 mm layer of air, while one thick blanket traps a single 5 mm layer. How much MORE air does the stack of three thin blankets trap than the one thick blanket? [Heat]

- **(A)** 4 mm
- **(B)** 11 mm
- **(C)** 2 mm
- **(D)** 1 mm
- **(E)** 5 mm
- **(F)** 3 mm
- **(G)** 6 mm

36. When a non-square rectangle is turned through one complete revolution, how many times does it look exactly the same as it started? [Symmetry]

- **(A)** 4
- **(B)** 2
- **(C)** 0
- **(D)** 180
- **(E)** 3
- **(F)** 1

37. A tall tree moves 1 litre of water up its xylem each hour during 11 daytime hours, but only one-quarter as fast through the 13 night hours. How much water rises in the whole 24 hours? [Transportation in Animals & Plants]

- **(A)** 3.25 L
- **(B)** 24 L
- **(C)** 11 L
- **(D)** 14.25 L
- **(E)** 14.5 L
- **(F)** 13.25 L
- **(G)** 17.25 L

38. Five plants have heights 12 cm, 15 cm, an unknown height, 20 cm and 18 cm. If the mean height is 16 cm, what is the missing height? [Data Handling]

- **(A)** 18 cm
- **(B)** 13 cm
- **(C)** 17 cm
- **(D)** 16 cm
- **(E)** 15 cm
- **(F)** 80 cm
- **(G)** 14 cm

39. A solar cooker's temperature rises from 30 C to 100 C in 35 minutes by absorbing the Sun's radiation. What is the average rise in temperature per minute? [Heat]

- **(A)** 2.85 C/minute
- **(B)** 3.5 C/minute
- **(C)** 100 C/minute
- **(D)** 2 C/minute
- **(E)** 0.5 C/minute
- **(F)** 1.5 C/minute

40. How many lines of symmetry does an ordinary parallelogram (not a rectangle or rhombus) have? [Symmetry]

- **(A)** 8
- **(B)** 0
- **(C)** 1
- **(D)** 3
- **(E)** 6
- **(F)** 2

41. A leaf makes 18 mg of sugar. The phloem sends two-thirds of it down to the roots and the rest up to a growing fruit. How much sugar reaches the fruit? [Transportation in Animals & Plants]

- **(A)** 18 mg
- **(B)** 6 mg
- **(C)** 9 mg
- **(D)** 8 mg
- **(E)** 12 mg
- **(F)** 3 mg
- **(G)** 4 mg

42. On a day the highest temperature was 41 C and the lowest was -3 C. What was the temperature range? [Data Handling]

- **(A)** -44 C
- **(B)** 41 C
- **(C)** 38 C
- **(D)** 48 C
- **(E)** 40 C
- **(F)** 44 C

43. A fever chart shows four readings: 38 C, 39.5 C, 40 C and 38.5 C. What is the mean (average) of these temperatures? [Heat]

- **(A)** 39 C
- **(B)** 39.5 C
- **(C)** 41 C
- **(D)** 156 C
- **(E)** 38 C
- **(F)** 40 C
- **(G)** 38.5 C

44. What is the smallest angle of rotational symmetry of a regular octagon? [Symmetry]

- **(A)** 60 degrees
- **(B)** 8 degrees
- **(C)** 30 degrees
- **(D)** 45 degrees
- **(E)** 90 degrees
- **(F)** 40 degrees

45. A model heart beats 75 times a minute. How many times does it beat in 2 hours? [Transportation in Animals & Plants]

- **(A)** 9000
- **(B)** 150
- **(C)** 18000
- **(D)** 9075
- **(E)** 900
- **(F)** 4500

46. What is the mode of the data set: 10, 10, 20, 30, 20, 20, 10, 10? [Data Handling]

- **(A)** 4
- **(B)** 13
- **(C)** 20
- **(D)** 10
- **(E)** 30
- **(F)** 16.25
- **(G)** 15

47. Heat travels 1 cm along an iron rod in 2 seconds, but along a wooden rod the same 1 cm takes 18 seconds. How many times faster does heat travel in iron than in wood? [Heat]

- **(A)** 18 times
- **(B)** 20 times
- **(C)** 9 times
- **(D)** 36 times
- **(E)** 8 times
- **(F)** 2 times

48. A regular figure has rotational symmetry of order 5, and (being regular) the same number of lines of symmetry. What is the sum of its lines of symmetry and its order of rotation? [Symmetry]

- **(A)** 15
- **(B)** 10
- **(C)** 8
- **(D)** 25
- **(E)** 72
- **(F)** 12

49. A doctor counts the pulse over six back-to-back 10-second intervals and gets 12, 13, 12, 14, 13 and 12 beats. What is the pulse rate per minute? [Transportation in Animals & Plants]

- **(A)** 760
- **(B)** 12
- **(C)** 456
- **(D)** 12.67
- **(E)** 76
- **(F)** 152

50. Class A has 20 students with a mean score of 60. Class B has 30 students with a mean score of 70. What is the mean score of all 50 students together? [Data Handling]

- **(A)** 65
- **(B)** 67
- **(C)** 66
- **(D)** 130
- **(E)** 68
- **(F)** 64
- **(G)** 3300

51. On a hot afternoon at the beach, the land heats up faster than the sea, so the air over the land rises. Which statement is correct? [Heat]

- **(A)** No breeze forms because both heat equally.
- **(B)** A land breeze blows during the hot afternoon.
- **(C)** Air over the sea rises first in the afternoon.
- **(D)** Cool air moves from land to sea as a land breeze.
- **(E)** Cool air moves from sea to land as a sea breeze.
- **(F)** Warm air sinks down over the land.
- **(G)** A sea breeze blows only at night.

52. The point (3, 5) is reflected in the y-axis. What is the sum of the coordinates of its image? [Symmetry]

- **(A)** 8
- **(B)** -2
- **(C)** 3
- **(D)** 5
- **(E)** -8
- **(F)** 15
- **(G)** 2

53. Healthy kidneys filter about 180 litres of fluid a day, but 99% of it is taken back into the blood. How much leaves the body as urine? [Transportation in Animals & Plants]

- **(A)** 18 L
- **(B)** 178.2 L
- **(C)** 1.78 L
- **(D)** 0.18 L
- **(E)** 1.8 L
- **(F)** 180 L
- **(G)** 17.82 L

54. A spinner has 8 equal sections numbered 1 to 8. What is the probability of landing on a multiple of 3? [Data Handling]

- **(A)** 3/8
- **(B)** 3/4
- **(C)** 1/4
- **(D)** 2/3
- **(E)** 1/8
- **(F)** 1/3
- **(G)** 1/2

55. A black tank and a white tank each hold 50 litres of water at 25 C in the sun. The black tank warms 0.4 C per minute and the white tank 0.15 C per minute. After 20 minutes, what is the temperature difference between them? [Heat]

- **(A)** 0.25 C
- **(B)** 13 C
- **(C)** 11 C
- **(D)** 3 C
- **(E)** 8 C
- **(F)** 5 C

56. The point (4, -7) is reflected in the x-axis. What is the product of the coordinates of its image? [Symmetry]

- **(A)** -11
- **(B)** -28
- **(C)** 3
- **(D)** 47
- **(E)** 11
- **(F)** 28
- **(G)** -3

57. A leaf has 300 stomata in every square millimetre. If the leaf surface is 40 square millimetres, how many stomata does it have? [Transportation in Animals & Plants]

- **(A)** 1,200
- **(B)** 12,000
- **(C)** 7,500
- **(D)** 340
- **(E)** 120,000
- **(F)** 12,300

58. In a frequency table, the mark 2 occurs 3 times, the mark 4 occurs 2 times, and the mark 6 occurs 5 times. What is the mean mark? [Data Handling]

- **(A)** 4.4
- **(B)** 12
- **(C)** 44
- **(D)** 4.8
- **(E)** 5.5
- **(F)** 4

59. In a thermometer the liquid column rises 2 mm for each 1 C rise in temperature. If the temperature goes up from 35 C to 37 C, how far does the column rise? [Heat]

- **(A)** 70 mm
- **(B)** 2 mm
- **(C)** 4 mm
- **(D)** 8 mm
- **(E)** 6 mm
- **(F)** 37 mm
- **(G)** 74 mm

60. Counting every diameter through its centre, how many fold lines map a circle exactly onto itself? [Symmetry]

- **(A)** 4
- **(B)** 1
- **(C)** infinite (countless)
- **(D)** 360
- **(E)** 0
- **(F)** 2

61. At night, root pressure pushes sap up a stem at 6 cm per hour. Over 9 night hours, how far up does the sap travel? [Transportation in Animals & Plants]

- **(A)** 63 cm
- **(B)** 54 cm
- **(C)** 45 cm
- **(D)** 540 cm
- **(E)** 15 cm
- **(F)** 48 cm
- **(G)** 6 cm

62. A bar chart shows monthly rainfall: 30 mm, 45 mm, 60 mm and 25 mm. By how much does the highest month exceed the mean rainfall? [Data Handling]

- **(A)** 20 mm
- **(B)** 60 mm
- **(C)** 15 mm
- **(D)** 160 mm
- **(E)** 35 mm
- **(F)** 25 mm
- **(G)** 40 mm

63. A science class sorts 6 everyday examples of heat flow: 2 are conduction, 3 are convection and 1 is radiation. What fraction of the examples are convection? [Heat]

- **(A)** 1/4
- **(B)** 2/6
- **(C)** 2/3
- **(D)** 3/2
- **(E)** 1/3
- **(F)** 1/6
- **(G)** 1/2

64. For an isosceles (but not equilateral) triangle, multiply its number of lines of symmetry by its order of rotational symmetry. What is the product? [Symmetry]

- **(A)** 9
- **(B)** 1
- **(C)** 2
- **(D)** 3
- **(E)** 6
- **(F)** 0
- **(G)** 4

65. Through a stethoscope a doctor hears 220 'lub-dub' sounds in 2.5 minutes. Each lub-dub is one heartbeat. What is the heart rate per minute? [Transportation in Animals & Plants]

- **(A)** 220
- **(B)** 73.3
- **(C)** 44
- **(D)** 110
- **(E)** 92
- **(F)** 550
- **(G)** 88

66. Find the median of the nine values: 7, 11, 11, 14, 18, 21, 25, 30, 33. [Data Handling]

- **(A)** 20
- **(B)** 21
- **(C)** 11
- **(D)** 19.78
- **(E)** 17
- **(F)** 14
- **(G)** 18

67. A cup of hot milk at 90 C cools at a steady 5 C every minute. What is its temperature after 8 minutes? [Heat]

- **(A)** 55 C
- **(B)** 30 C
- **(C)** 40 C
- **(D)** 50 C
- **(E)** 58 C
- **(F)** 45 C
- **(G)** 85 C

68. How many lines of symmetry does a kite have? [Symmetry]

- **(A)** 6
- **(B)** 2
- **(C)** 3
- **(D)** 4
- **(E)** 0
- **(F)** 1

69. A person donates 450 mL of blood from a total of 5 litres. What fraction of the blood is donated, in simplest form? [Transportation in Animals & Plants]

- **(A)** 9/100
- **(B)** 45/100
- **(C)** 1/10
- **(D)** 90/100
- **(E)** 9/50
- **(F)** 9/1000

70. From cards labelled 1 up to 20, one card is picked at random. Find the chance that its number is both even and larger than 10. [Data Handling]

- **(A)** 1/10
- **(B)** 3/10
- **(C)** 3/4
- **(D)** 1/4
- **(E)** 1/5
- **(F)** 1/2

71. A steel spoon and a plastic spoon are dipped in tea at 70 C in a room at 30 C. The steel handle warms up to 60 C, while the plastic handle stays near 30 C. What is the temperature difference between the two handles? [Heat]

- **(A)** 20 C
- **(B)** 30 C
- **(C)** 100 C
- **(D)** 40 C
- **(E)** 70 C
- **(F)** 60 C
- **(G)** 10 C

72. A windmill has 4 identical blades, all bent the same way, so it has rotational symmetry of order 4 but no lines of symmetry. What is its smallest angle of rotation that maps it onto itself? [Symmetry]

- **(A)** 72 degrees
- **(B)** 45 degrees
- **(C)** 180 degrees
- **(D)** 4 degrees
- **(E)** 120 degrees
- **(F)** 90 degrees
- **(G)** 360 degrees

73. Over 4 hours of play a child loses 600 mL of sweat. The loss runs at a steady rate for the first 2 hours, then DOUBLES for the last 2 hours. How much sweat is lost in the last 2 hours? [Transportation in Animals & Plants]

- **(A)** 300 mL
- **(B)** 200 mL
- **(C)** 100 mL
- **(D)** 500 mL
- **(E)** 600 mL
- **(F)** 400 mL
- **(G)** 150 mL

74. The mean of 7 numbers is 30. One of the numbers, which equals 30, is removed. What is the mean of the remaining 6 numbers? [Data Handling]

- **(A)** 5
- **(B)** 30
- **(C)** 25
- **(D)** 35
- **(E)** 36
- **(F)** 210
- **(G)** 180

75. A glass greenhouse traps heat. Its inside temperature rises 3 C per hour for 4 hours during the day, then loses 2 C per hour for 3 hours at night. What is the net change in temperature from the start? [Heat]

- **(A)** +9 C
- **(B)** -6 C
- **(C)** +6 C
- **(D)** +12 C
- **(E)** 0 C
- **(F)** +18 C
- **(G)** +2 C

76. A regular pentagon has a smallest rotational-symmetry angle of 72 degrees. Through how many degrees has it turned after 3 such smallest rotations? [Symmetry]

- **(A)** 180 degrees
- **(B)** 288 degrees
- **(C)** 108 degrees
- **(D)** 360 degrees
- **(E)** 144 degrees
- **(F)** 72 degrees
- **(G)** 216 degrees

77. A dialysis machine cleans 250 mL of blood every minute for 4 hours. How many litres of blood does it process in all? [Transportation in Animals & Plants]

- **(A)** 100 L
- **(B)** 60 L
- **(C)** 600 L
- **(D)** 6 L
- **(E)** 15 L
- **(F)** 1 L

78. A double bar graph shows that 4 classes had 240 boys in total and 200 girls in total. Treating all 8 bars as data, what is the mean number of students per bar? [Data Handling]

- **(A)** 65
- **(B)** 60
- **(C)** 110
- **(D)** 27.5
- **(E)** 55
- **(F)** 50
- **(G)** 440

79. Over five days a city's noon temperatures were 31 C, 33 C, 30 C, 35 C and one more day. If the mean of all five days is 32 C, what was the fifth day's temperature? [Heat]

- **(A)** 160 C
- **(B)** 31 C
- **(C)** 34 C
- **(D)** 32 C
- **(E)** 30 C
- **(F)** 33 C
- **(G)** 29 C

80. For a rhombus (not a square), multiply the number of lines of symmetry by the order of rotational symmetry. What is the product? [Symmetry]

- **(A)** 8
- **(B)** 2
- **(C)** 4
- **(D)** 0
- **(E)** 1
- **(F)** 16
- **(G)** 6

81. Right after a run a girl's pulse is 150 beats per minute. Each minute of rest it drops by one-fifth of the rate at the start of that minute. What is her pulse after 2 minutes of rest? [Transportation in Animals & Plants]

- **(A)** 108
- **(B)** 30
- **(C)** 144
- **(D)** 96
- **(E)** 120
- **(F)** 60

82. Over 30 days, 18 were sunny, 9 were cloudy and 3 were rainy. If one day is picked at random, what is the probability that it was NOT rainy? [Data Handling]

- **(A)** 1/10
- **(B)** 3/5
- **(C)** 2/3
- **(D)** 3/10
- **(E)** 4/5
- **(F)** 1/3
- **(G)** 9/10

83. A vacuum flask keeps coffee from cooling fast, losing only 1.5 C per hour. If coffee starts at 85 C, what is its temperature after 6 hours? [Heat]

- **(A)** 85 C
- **(B)** 74 C
- **(C)** 9 C
- **(D)** 76 C
- **(E)** 79 C
- **(F)** 70 C

84. What is the order of rotational symmetry of the letter Z? [Symmetry]

- **(A)** 3
- **(B)** 1
- **(C)** 4
- **(D)** 2
- **(E)** 180
- **(F)** infinite

85. Blood returns to the heart at about 5 litres per minute. How many litres flow back to the heart in one full day? [Transportation in Animals & Plants]

- **(A)** 3600 L
- **(B)** 300 L
- **(C)** 1440 L
- **(D)** 5 L
- **(E)** 7200 L
- **(F)** 120 L

86. What is the mean of all the integers from -4 to 6, including both? [Data Handling]

- **(A)** 6
- **(B)** 2
- **(C)** 11
- **(D)** 1.1
- **(E)** 0
- **(F)** 1
- **(G)** -1

87. Land heats up about 3 times as fast as water in the same sunshine. If the water's temperature rises 4 C, by how much does the land's temperature rise? [Heat]

- **(A)** 12 C
- **(B)** 36 C
- **(C)** 9 C
- **(D)** 1.33 C
- **(E)** 7 C
- **(F)** 4 C

88. A regular polygon has a smallest rotational-symmetry angle of 30 degrees. What is the sum of its number of sides and its number of lines of symmetry? [Symmetry]

- **(A)** 20
- **(B)** 12
- **(C)** 60
- **(D)** 24
- **(E)** 15
- **(F)** 36
- **(G)** 30

89. Of 250 mL of water taken in by a plant, 98% is lost by transpiration and the rest is actually used in the plant's work. How many millilitres are used? [Transportation in Animals & Plants]

- **(A)** 245 mL
- **(B)** 2.5 mL
- **(C)** 248 mL
- **(D)** 5 mL
- **(E)** 50 mL
- **(F)** 0.5 mL

90. Shoe sizes recorded were 5, 6, 6, 7, 8, 8, 8 and 9. What is the mean shoe size? [Data Handling]

- **(A)** 8.125
- **(B)** 57
- **(C)** 8
- **(D)** 6.5
- **(E)** 7.5
- **(F)** 7.125

91. Two 1-metre rods are heated equally. The aluminium rod grows 2.4 mm and the iron rod grows 1.2 mm. By what percentage does aluminium expand MORE than iron? [Heat]

- **(A)** 50%
- **(B)** 200%
- **(C)** 20%
- **(D)** 12%
- **(E)** 100%
- **(F)** 2%
- **(G)** 120%

92. The point (2, 9) is reflected across the line y = x. In its image, what is the new x-coordinate minus the new y-coordinate? [Symmetry]

- **(A)** -7
- **(B)** -11
- **(C)** 0
- **(D)** 7
- **(E)** 18
- **(F)** 11

93. A newborn baby's heart beats about 140 times a minute, while an adult's beats about 70 times a minute. By what percentage is the newborn's rate higher than the adult's? [Transportation in Animals & Plants]

- **(A)** 200%
- **(B)** 70%
- **(C)** 140%
- **(D)** 25%
- **(E)** 100%
- **(F)** 50%
- **(G)** 75%

94. Find the median of: 2.5, 3.0, 3.5, 4.0, 4.5, 5.0. [Data Handling]

- **(A)** 4.25
- **(B)** 3.25
- **(C)** 3.5
- **(D)** 3.67
- **(E)** 4.0
- **(F)** 22.5
- **(G)** 3.75

95. 3 litres of water at 80 C is mixed with 3 litres of water at 20 C in an insulated jug. What is the final temperature of the mixture (equal amounts mix to the middle)? [Heat]

- **(A)** 40 C
- **(B)** 100 C
- **(C)** 30 C
- **(D)** 60 C
- **(E)** 55 C
- **(F)** 45 C
- **(G)** 50 C

96. The minute hand of a clock moves from the 12 to the 3, turning a quarter circle. If a regular figure has this as its smallest angle of rotational symmetry, what is its order of rotation? [Symmetry]

- **(A)** 90
- **(B)** 3
- **(C)** 2
- **(D)** 8
- **(E)** 12
- **(F)** 4
- **(G)** 6

97. A leaf has 12 small veins. Three-quarters of them carry mostly water-rich xylem flow and the rest carry mostly food-rich phloem flow. How many veins carry mostly phloem flow? [Transportation in Animals & Plants]

- **(A)** 12
- **(B)** 3
- **(C)** 2
- **(D)** 8
- **(E)** 6
- **(F)** 9
- **(G)** 4

98. A bar graph shows a shop sold 200 items in a day: 25% were type A, 35% were type B, and the rest were type C. How many type C items were sold? [Data Handling]

- **(A)** 60
- **(B)** 120
- **(C)** 100
- **(D)** 70
- **(E)** 40
- **(F)** 80

99. A solar panel's temperature rises 6 C every 10 minutes when it faces the Sun, but only 6 C every 30 minutes when tilted away. To gain 18 C in all, how many more minutes does the tilted panel need than the facing one? [Heat]

- **(A)** 120 minutes
- **(B)** 60 minutes
- **(C)** 18 minutes
- **(D)** 45 minutes
- **(E)** 20 minutes
- **(F)** 90 minutes

100. A semicircle has how many lines of symmetry, and what is its order of rotational symmetry? Give the sum of these two numbers. [Symmetry]

- **(A)** 1
- **(B)** 6
- **(C)** 0
- **(D)** 2
- **(E)** 3
- **(F)** 4